Engineering Change Notice - Gearbox Flange Joint and Sealant Bead

Document no.	ECN-2026-044	**Revision**	A
Document type	Engineering Change	**Status**	Released
Effective date	2026-06-01	**Owner**	Product Engineering
Classification	Internal		

1 Reason for change

Customer complaint 8D-2026-017 (oil leakage at the gearbox flange). The joint analysis showed that the clamping force at 40 Nm was too low after thermal cycling for the sealant bead in use.

2 Description of change

Item	Old	New
Torque M10 flange bolts	40 Nm +/- 4 Nm	45 Nm +/- 4.5 Nm
Sealant bead width	2.5 mm	3.0 mm
Sealant open time	15 min	10 min
Curing time before leak test	20 min	30 min
Bead inspection	visual by operator	camera inspection plus operator confirmation

3 Validation

The 45 Nm torque was validated by the bolt calculation according to VDI 2230 and by 500 units with thermal cycling (-40 to +120 degrees C, 50 cycles). The M10 bolt strength class 10.9 is not changed. Surface pressure under the bolt head stays below the limit of the aluminium housing.

4 Affected documents

WI-ASM-014 (Rev B -> C), WI-PNT-005 (Rev C -> D), PFMEA dispensing, control plan L2.

5 Effectivity

Effective from 2026-06-01, starting with serial number ED210-2606-00001. Units built before this date with 40 Nm do not need to be reworked if they passed the helium sniff test during containment.

6 Approval

Approved by product engineering, quality and production on 2026-05-27. Customer informed and approved the change on 2026-05-29.